EMPLOYEE PAYROLL SYSTEM

Python Programming Case Study

1. Problem Statement

Organizations need to calculate employee salaries accurately by considering basic pay, allowances, overtime and incentives. They must also calculate tax and provident fund (PF) deductions to determine the employee's net salary.

2. Objective

To develop a Python program that accepts employee salary details, calculates gross salary, applies progressive tax rates, calculates PF and total deductions, and displays the final net salary in a salary statement.

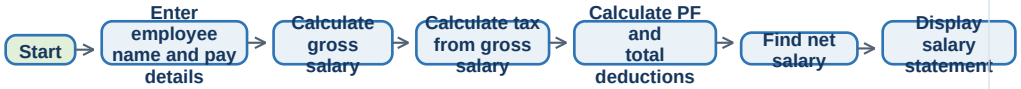
3. Assumptions

- All salary inputs are entered as numeric, non-negative values.
- Gross salary = basic pay + allowances + overtime + incentives.
- Tax is 5% when gross salary is below 10,000; 10% when it is from 10,000 to below 20,000; and 15% when it is 20,000 or more.
- PF is 12% of basic pay.
- Total deductions = tax + PF; net salary = gross salary – total deductions.

4. Algorithm

1. Start the program and input the employee's name.
2. Read basic pay, allowances, overtime and incentives.
3. Calculate gross salary by adding all four amounts.
4. Calculate tax using the applicable gross-salary range: 5%, 10% or 15%.
5. Calculate PF as 12% of basic pay.
6. Add tax and PF to get total deductions.
7. Subtract total deductions from gross salary to get net salary.
8. Display the salary statement and stop.

5. Flowchart



6. Pseudocode

BEGIN
INPUT name, basic, allowance, overtime, incentive
gross ← basic + allowance + overtime + incentive
IF gross < 10000: tax ← gross × 0.05
ELSE IF gross < 20000: tax ← gross × 0.10
ELSE: tax ← gross × 0.15
pf ← basic × 0.12
total_deduction ← tax + pf
net ← gross – total_deduction
DISPLAY name, basic, gross, tax, pf, total_deduction, net
END

7. Python Source Code and Output

Original screenshot supplied for this case study

```python
# Employee Payroll System

name = input("Enter Employee Name: ")

basic = float(input("Enter Basic Pay: "))
allowance = float(input("Enter Allowances: "))
overtime = float(input("Enter Overtime: "))
incentive = float(input("Enter Incentives: "))

# Gross Salary
gross = basic + allowance + overtime + incentive

# Progressive Deduction
if gross <= 10000:
    tax = gross * 0.05
elif gross <= 20000:
    tax = gross * 0.10
else:
    tax = gross * 0.15

pf = basic * 0.12
total = tax + pf

net = gross - total

# Salary Statement
print("\n--- SALARY STATEMENT ---")
print("Employee:", name)
print("Basic Pay:", basic)
print("Gross Salary:", gross)
print("Tax:", tax)
print("PF:", pf)
print("Total Deduction:", total)
print("Net Salary:", net)
```

```
Problems  Output  Debug Console  Terminal  Ports

Enter Incentives: 7

--- SALARY STATEMENT ---
Employee: ganesh
Basic Pay: 600.0
Gross Salary: 1012.0
Tax: 50.6
PF: 72.0
Total Deduction: 122.6
Net Salary: 889.4
PS C:\Users\jakku\OneDrive\Desktop\GANII>
```

8. Conclusion

The Employee Payroll System demonstrates Python input, arithmetic operations and conditional statements. It calculates gross salary, determines tax according to the given salary slabs, computes PF and total deductions, and displays the employee's net salary. This helps make basic payroll calculations organized and consistent.